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Competitive Coevolutionary PPO for Cryptocurrency Trading (Team 100)

23CSE399 – Project Phase 1 (Panel Review 1)

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1 Problem Definition & Originality

Problem: Standard DRL agents trained on static historical data converge on a single locally optimal strategy (*strategy collapse*). These identical, overfitted agents fail simultaneously in unseen market regimes. We address the need to explicitly distinguish between performance convergence and behavioural convergence in cryptocurrency trading.

Originality (CC-PPO Framework): We propose **CC-PPO**, a competitive coevolutionary wrapper around PPO with four novel components: (1) **Dominance-Based Fitness** — agents are ranked by the fraction of opponents they beat, not absolute profit; (2) **Strategy Diversity Index (SDI)** — a diversity bonus via mean KL divergence of action distributions, penalizing redundant trades; (3) **Adaptive Opponent Sampling** — EMA-tracked win-rates oversample difficult opponents to patch weaknesses; (4) **MAP-Elites Archiving** — preserves past elites in a behavioural grid, structurally guaranteeing diversity.

2 Literature Survey & Gap Analysis

Paper	Key Findings	Gap Identified
Alexandrov et al. [1]	Arms-race dynamics in multi-agent trading; need for better analytic tools.	Limited to rule-based agents; calls for expressive DRL architectures.
Sun et al. [2]	Population-based DRL achieves high backtest returns.	Evaluates against fixed data; conflates performance and behavioural convergence.
Schulman et al. [3]	PPO's clipped surrogate objective ensures stable policy updates.	Prone to local optima on static datasets without diversity pressure.
Pugh et al. [4]	MAP-Elites grid guarantees diversity via behavioural niches.	Not yet applied to deep RL trading to prevent strategy collapse.
Watson & Pollack [5]	Populations without diversity mechanisms suffer pathological cycling.	Prior RL trading relies on emergent rather than structural diversity.
Jaderberg et al. [6]	PBT optimizes hyperparameters via exploit/explore cycles.	Used for isolated tuning, not adversarial evolutionary arms races.

Key Gaps: Static fitness signals make behavioural convergence inevitable; standard DRL achieves high returns but near-zero diversity; existing coevolutionary models use simple rules while deep RL models lack coevolutionary diversity [1, 2].

3 Proposed Methodology (Technical Knowledge)

PPO Base Learner: Each agent uses PPO with an Actor-Critic MLP (64×64). The clipped surrogate objective $L^{\text{CLIP}}(\theta) = \mathbb{E}_t[\min(r_t(\theta)\hat{A}_t,\ \text{clip}(r_t(\theta), 1-\varepsilon, 1+\varepsilon)\hat{A}_t)]$ prevents destructive weight updates during volatile price swings [3]. Input shape is (20×7) — 20 hourly BTC/USDT candles processed into 7 features: Log Return, Rolling Mean Return, Rolling Volatility, RSI(14), Drawdown Proxy, Z-Score, and Portfolio Position. Output is Discrete(3): Hold, Buy, or Sell via softmax.

Coevolutionary Meta-Layer: The combined fitness $fitness(i) = dominance_fitness(i) + \lambda \times diversity_bonus(i)$ drives the evolutionary loop: Adaptive Opponent Sampling $\rightarrow$ Competitive Evaluation $\rightarrow$ SDI & Fitness Calculation $\rightarrow$ Tournament Selection $\rightarrow$ PBT-Style Reproduction (exploit top weights, explore hyperparameters) $\rightarrow$ MAP-Elites Grid Update.

Data: Binance Public API, BTC/USDT, 1-hour candles, close price only. \$10,000 simulated capital with 0.1% fee per trade. Strict temporal train/test split (no lookahead bias).

Preliminary Results: A single-agent PPO baseline on 5,000 hours of data yielded -4.34% return vs. Buy & Hold at $+6.78\%$, confirming PPO's sensitivity to transaction fees. A population of 4 agents (seeds 42–45) produced a Diversity Score of only 0.1 at realistic fees — total collapse to a "Do-Nothing" policy. This empirically validates that random seeds cannot prevent strategy collapse; explicit coevolutionary diversity pressure is required.

4 Review Comments

S.No	Review Comment
1	Find a validation strategy for the project.

5 Current Status & Plan

Status: Literature survey complete; strategy collapse problem formalized; CC-PPO framework designed (Dominance Fitness, SDI, Adaptive Opponent Sampling, MAP-Elites); single-agent PPO baseline implemented on real Binance data; strategy collapse empirically demonstrated.

Next Steps: Find a validation strategy and also understand the inferences of PPO output.

References

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